PAPER ID-310890

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2023-24
STATISTICAL COMPUTING

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

Q no.	Question	Marks
a.	Discuss the significance of measures of dispersion in a dataset. Provide examples of two measures of dispersion and explain how they differ.	2
b.	Define the concept of "mean" as a measure of central tendency. How is it calculated, and what are its advantages and limitations?	2
c.	Explain the concept of correlation and its significance in statistical analysis. Provide an example of a positive and a negative correlation.	2
d.	Outline the steps involved in conducting an inference procedure for a correlation coefficient. Why is it important in statistical analysis?	2
e.	Differentiate between bivariate correlation and simple correlation. Provide an example scenario where bivariate correlation would be more appropriate.	2
f.	Explain the concept of linear regression. How is the regression line determined, and what does the slope of the line represent in a regression analysis?	2
g.	Describe the key differences between simple linear regression and multiple regression. Why might a researcher choose to use multiple regression?	2
h.	Compute the correlation coefficient for the following paired data: X (hours of study) - 10, 15, 20, 25; Y (exam scores) - 60, 75, 80, 90.	2
i.	Given the dataset: X - 2, 4, 6, 8; Y - 5, 8, 11, 14, find the equation of the regression line (Y on X).	2
j.	In a deck of 52 cards, what is the probability of drawing a queen given that the card drawn is a face card?	2

SECTION B

2. Attempt any *three* of the following:

a.	Apply singular value decomposition (SVD) to a given matrix B. Discuss the significance of the singular values and how they contribute to linear dimension reduction.	10
b.	Conduct a multiple regression analysis using a dataset with two independent variables. Interpret the coefficients and assess the overall fit of the model.	10
c.	In a study comparing two teaching methods, Group A has 15 students taught with Method 1, and Group B has 20 students taught with Method 2. The exam scores for both groups are given. Use a randomization test to determine if there is a significant difference in the mean scores between the two groups.	10
d.	(i) Define principal component analysis (PCA) and its purpose in linear dimension reduction. Discuss the steps involved in PCA. (ii) Apply PCA to a dataset with five variables. Interpret the results and explain how PCA aids in dimensionality reduction.	10
e.	A random sample of 50 students is selected from a population. The average hours of study per week for the sample is 15 hours, with a standard deviation of 3 hours. Calculate a 95% confidence interval for the mean hours of study for the population.	10

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SECTION C

3. Attempt any *one* part of the following:

a.	Perform a Monte Carlo simulation to test the hypothesis that the mean of a population is 50, based on a sample of 30 observations with a known standard deviation of 10. Use 1000 simulated samples and a significance level of 0.05.	10
b.	Define Markov chains and explain their role in Markov Chain Monte Carlo (McMC) methods. Discuss the importance of convergence in McMC simulations.	10

4. Attempt any *one* part of the following:

a.	Explain how Monte Carlo simulations can be used for hypothesis testing. Discuss the advantages of Monte Carlo hypothesis testing over traditional methods.	10
b.	Explain the concept of the jackknife resampling method. Apply the jackknife to estimate the variance of the mean for a dataset with 30 observations.	10

5. Attempt any *one* part of the following:

a.	Perform a permutation test to compare the means of two independent groups (Group A: 25 observations, Group B: 30 observations). The observed difference in means is 2.5, and the permutation distribution yields differences of -1.8, -0.5, 1.0, ...	10
b.	Use the jackknife resampling method to estimate the bias and variance of the mean for a dataset with 40 observations. Present the jackknife-estimated mean, bias, and variance.	10

6. Attempt any *one* part of the following:

a.	Perform 5-fold cross-validation on a linear regression model. The dataset has 80 observations, and the mean squared error for each fold is as follows: 12, 15, 10, 18, 14. Compute the average mean squared error across the folds.	10
b.	Briefly discuss the history and origin of the R programming language. Highlight key milestones and contributors.	10

7. Attempt any *one* part of the following:

a.	Save a vector of numbers (e.g., c(2, 4, 6, 8, 10)) to the R workspace. Show how to inspect the variables in the workspace and display their values.	10
b.	Define a vector with the elements 1 to 5. Create a matrix with 3 rows and 2 columns using the vector. Display both the vector and the matrix.	10

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